FireLink 🔥 → 💧 🚒

**AI-powered wildfire awareness —
linking fire incidents to nearby water sources and fire stations**

By Harshavardhana Naganagoudar

Problem: The Growing Wildfire Challenge

- Wildfires are becoming more frequent and intense due to climate change.
- Rapid response is critical — but responders need context: Where is the fire, and what resources are nearby?
- Current tools often lack integration of fire data with real-time maps of water access and fire stations.
- There's a gap in awareness tools that combine live satellite data, infrastructure proximity, and simple interfaces.

Solution: FireLink - From Detection to Action

- FireLink is an AI-Agent that maps recent wildfire activity in near real time.
- It links fire locations to the **nearest water bodies** and **fire stations** using live geospatial data.
- Users can simply type natural language prompts like: "Show fires in California with brightness over 300 and confidence above 80"
- FireLink returns a clear, interactive map for situational awareness and planning.

Key Features of FireLink

🔥 Real-Time Wildfire Detection — Pulls satellite data from NASA FIRMS (MODIS/VIIRS)

💧 Water Source Mapping — Uses Google Earth Engine to locate historically persistent water bodies

🚒 Fire Station Proximity — Retrieves fire station data from OpenStreetMap in real-time

🧠 Natural Language Interface — AI agent parses user prompts like "fires in California with brightness > 300"

🗺️ Interactive Map Output — Visualizes fire spots with linked water and station locations on a live map

How FireLink Works

1️⃣ User submits a natural language query
e.g., "Show fires in California with brightness > 300"

2️⃣ AI agent parses intent and parameters
(location = California, brightness = 300, etc.)

3️⃣ FireLink fetches data from:
🔥 NASA FIRMS (fire hotspots)
💧 Google Earth Engine (water bodies)
🚒 OpenStreetMap (fire stations)

4️⃣ It maps:
- Fire location
- Nearest water source
- Nearest fire station

5️⃣ Returns a live, interactive map as HTML

Real-Life Use Cases

🚨 Emergency Response Planning

Quickly assess fire proximity to fire stations and water — essential for dispatch and strategy.

🌲 Forest & Wildlife Protection

Identify at-risk conservation areas where infrastructure is limited.

🧯 Infrastructure Readiness

Find gaps where fire-prone areas lack nearby water or response stations.

🌾 Agricultural Risk Assessment

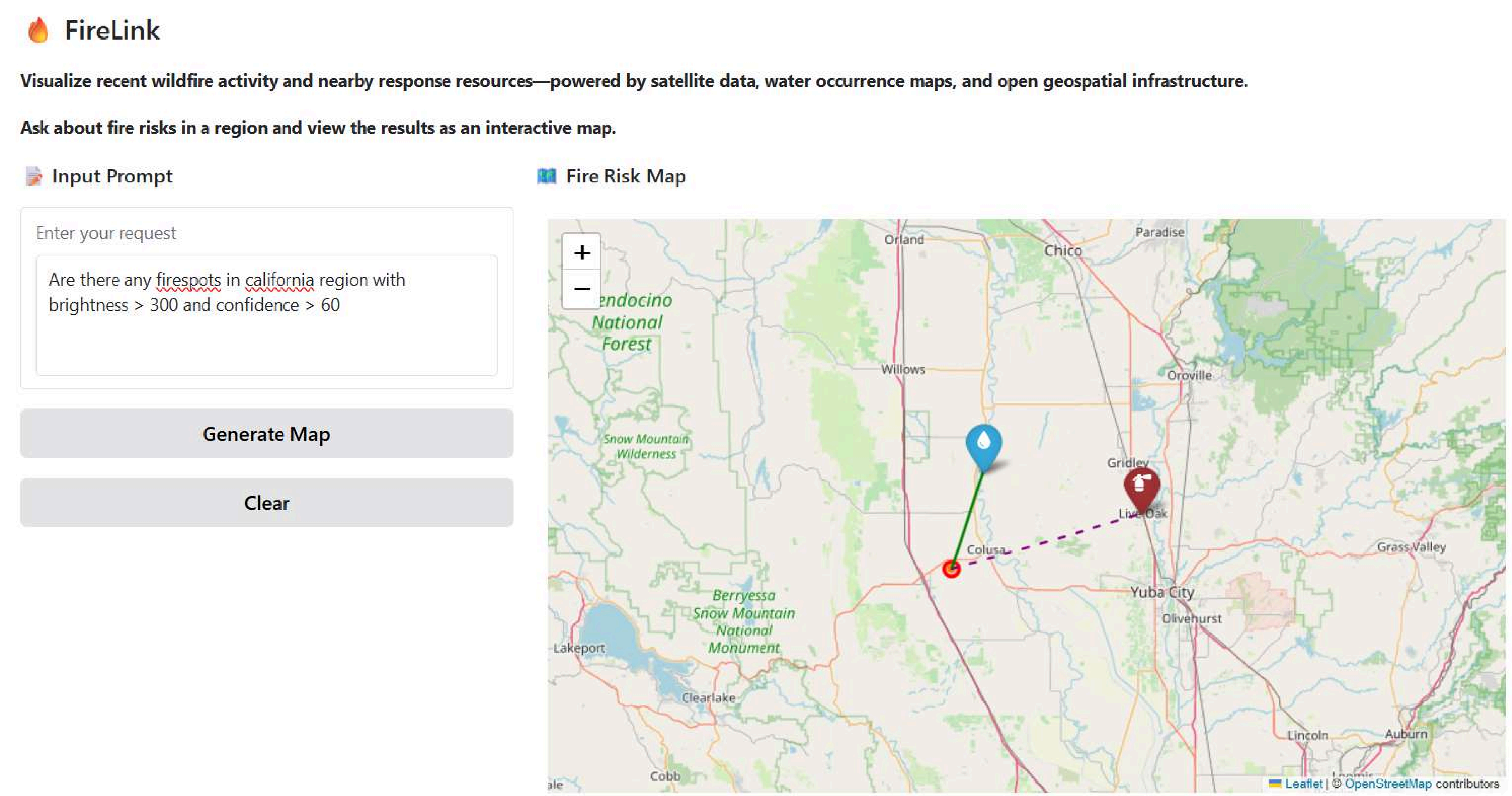
Support farmers in tracking wildfire activity near cropland or irrigation resources.

📢 Community Awareness & Education

Help rural communities visualize local fire risks and preparedness zones.

FireLink in Action

"FireLink identifies active fire spots and displays nearby water sources and fire stations on an interactive map." (App link: Try live demo)



GitHub repository: https://github.com/HarshavardhanaNaganagoudar/FireLink_AI_Agent

What Powers FireLink

LayerTools & Technologies

🔍 AI Agent - Mistral + LangGraph (tool-using agent for natural language

🔥 Fire Data - NASA FIRMS (MODIS/VIIRS satellite data)

💧 Water Data - JRC Global Surface Water via Google Earth Engine

🚒 Station Data - OpenStreetMap via Overpass API

🗺️ Mapping - Folium (interactive maps)

🧠 Backend - Python, Pandas, Requests, Geopy

💻 UI - Gradio (Blocks interface)

🚀 Deployment - Hugging Face Spaces

Why FireLink Matters

🚒 Faster, Smarter Response

Helps responders visualize infrastructure proximity in real-time for better decision-making.

🌍 Scalable Wildfire Awareness

Works globally using open geospatial data and natural language — no GIS experience required.

📢 Informed Communities

Makes fire risk data understandable and accessible to non-technical users and local residents.

🛠️ Infrastructure Planning

Reveals regions that lack water access or fire stations near recurring fire zones.

📊 Data-Driven Policy Support

Useful for research, reporting, and environmental strategy in climate-sensitive regions.

What’s Next for FireLink

✅ Phase 1: Current Release

- AI-powered map generation
- Near Real-time fire detection
- Water + station proximity mapping

🧭 Phase 2: Alert System

- Email or SMS notifications for new fire events near sensitive zones

📱 Phase 3: Mobile-Optimized Version

- Access FireLink on phones for use in the field

📡 Phase 4: Live Sensor Integration

- Ingest IoT or drone-based data for hyperlocal updates

🤝 Phase 5: API for Emergency Agencies

- Provide map and fire-risk data as an API for integration into government tools

Team & Acknowledgments

👤 Project Lead: Harshavardhana Naganagoudar
📍 Location: India
🔗 GitHub: https://github.com/HarshavardhanaNaganagoudar
📫 LinkedIn: https://www.linkedin.com/in/harshavardhana-naganagoudar-75901299/

🔧 Built With:
- NASA FIRMS API
- Google Earth Engine
- OpenStreetMap
- LangGraph + Mistral
- Gradio (Hugging Face Spaces)

🙏 Special Thanks:
- The open-source geospatial and AI community
- Earth observation & mapping datasets used

THANK YOU